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VIBROACOUSTIC DEVICES, SYSTEMS, AND METHODS FOR PROMOTING REST

Abstract

Vibroacoustic systems, devices, and methods for using the same are presently disclosed. An example method can include generating, using a vibroacoustic device having a vibroacoustic speaker and an acoustic speaker, a first combination of vibrations and acoustic noises, generating, using the vibroacoustic device, a transition combination of vibrations and acoustic noises, wherein the transition combination is a gradual transition of intensity of vibrations and volume of acoustic noises from the a first intensity and volume of the first combination to a second intensity and volume of a second combination of vibrations and acoustic noises, and generating, using the vibroacoustic device, the second combination of vibrations and acoustic noises.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/563,958, filed on Mar. 11, 2024, entitled "VIBROACOUSTIC DEVICES AND SYSTEMS FOR PROMOTING REST," and U.S. Provisional Patent Application No. 63/766,335, filed on Mar. 3, 2025, entitled "VIBROACOUSTIC DEVICES AND SYSTEMS FOR PROMOTING REST, both of which are expressly incorporated by reference herein in its entirety.

TECHNICAL FIELD
[0002] The subject disclosure relates generally to vibroacoustic devices. More particularly, the present disclosure relates to vibroacoustic devices for promoting sleep or rest.
BACKGROUND
[0003] Sleep is one of the most healthy and helpful methods of allowing our bodies to recoup and recover. It is universally needed, not only for all humans, but for animals as well. As society modernizes and the advent of non-stop distractions proliferates, the human mind becomes more distracted and cluttered. Thus, for millions of people in the U.S. and around the world, not just getting to bed, but being able to relax and fall asleep have become much more difficult.
[0004] Various drugs have been developed to induce rest and sleep. However, all of these drugs have certain side effects and run the risk of developing dependencies, as well as the risk of overdose.
[0005] For children, in particular, being able to promote rest and sleep, particularly with a nap or nighttime routine, is especially important, as such habits developed at childhood carry on into adulthood.
[0006] Not only parents, but all consumers, would benefit greatly in seeking non-medication techniques to induce rest or sleep. What is needed are devices and techniques that promote rest and sleep without the negative effects, dangers, and risks of drugs. Vibroacoustic devices are systems that produce or transmit vibrations and sound waves. They are commonly used in applications such as medical devices, industrial equipment, and musical instruments. These devices typically consist of a source of vibration (e.g., a speaker or a motor) and a shell or enclosure that transmits the vibrations to an external medium (e.g., air or water).
[0007] One of the key challenges in designing vibroacoustic devices is efficiently transmitting sound and vibration through the shell. The shell must be designed to optimize the transfer of energy from the source of vibration to the external medium, while minimizing energy loss or absorption. This can be achieved through various shell designs and materials, such as resonant cavities, acoustic lenses, and thin films.
BRIEF SUMMARY
[0008] The present subject disclosure presents a simplified summary of the subject disclosure in order to provide a basic understanding of some aspects thereof. This summary is not an extensive overview of the various embodiments of the subject disclosure. It is intended to neither identify key or critical elements of the subject disclosure nor delineate any scope thereof. The sole purpose of the subject summary is to present some concepts in a simplified form as a prelude to the more

detailed description that is presented hereinafter.
[0009] In one exemplary embodiment, the present subject disclosure is a vibroacoustic system. The vibroacoustic system can include a vibroacoustic device and control features. The vibroacoustic device can have a housing, a vibroacoustic speaker disposed in the housing, the vibroacoustic speaker configured to generate vibrations, and an acoustic speaker disposed in the housing, the acoustic speaker configured to generate acoustic sounds. The control features are configured to control the vibroacoustic device, wherein the control features cause the vibroacoustic device to generate coordinated vibrations and acoustic sounds.
[0010] In another exemplary embodiment, the present subject disclosure is a vibroacoustic device. The vibroacoustic device can include a housing; a vibroacoustic speaker disposed in the housing, the vibroacoustic speaker configured to generate vibrations; an acoustic speaker disposed in the housing, the acoustic speaker configured to generate acoustic sound; and a printed circuit board (PCB) in communication with the vibroacoustic speaker and the acoustic speaker, wherein the PCB causes the vibroacoustic speaker and the acoustic speaker to generate coordinated vibrations and acoustic sounds.
[0011] In another exemplary embodiment, the present subject disclosure is a method. The method can include generating, using a vibroacoustic device having a vibroacoustic speaker and an acoustic speaker, a first combination of vibrations and acoustic noises; generating, using the vibroacoustic device, a transition combination of vibrations and acoustic noises, wherein the transition combination is a gradual transition of intensity of vibrations and volume of acoustic noises from the a first intensity and volume of the first combination to a second intensity and volume of a second combination of vibrations and acoustic noises; and generating, using the vibroacoustic device, the second combination of vibrations and acoustic noises.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] Various exemplary embodiments of this disclosure will be described in detail. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the subject disclosure and technical data supporting those embodiments, and together with the written description, serve to explain certain principles of the subject disclosure. With reference to the following figures, wherein:
[0013] FIG. **1** illustrates an example vibroacoustic system, according to some aspects of the present disclosure.
[0014] FIG. **2** illustrates an example vibroacoustic device and remote control, according to some aspects of the present disclosure.
[0015] FIG. **3** is an exploded view of an example vibroacoustic device, according to some aspects of the present disclosure.
[0016] FIG. **4** is a top view of an example vibroacoustic device, according to some aspects of the present disclosure.
[0017] FIG. **5** is a bottom view of an example vibroacoustic device, according to some aspects of the present disclosure.
[0018] FIG. **6** is a lower perspective view of an example vibroacoustic device, according to some aspects of the present disclosure.
[0019] FIG. **7** is an upper perspective view of an example vibroacoustic device, according to some aspects of the present disclosure.
[0020] FIG. **8** is a side view of an example vibroacoustic device, according to some aspects of the present disclosure.
[0021] FIG. **9** is a cross-section view through section A-A in FIG. **5** of an example vibroacoustic

device, according to some aspects of the present disclosure.

[0022] FIG. **10** illustrates an example vibroacoustic device attached to a crib, according to some aspects of the present disclosure.

[0023] FIG. **11** illustrates attachment of an example vibroacoustic device to a crib, according to some aspects of the present disclosure.

[0024] FIG. **12** illustrates an example vibroacoustic device attached to a frame of a crib, according to some aspects of the present disclosure.

[0025] FIG. **13** illustrates attachment of an example vibroacoustic device to a frame of a crib, according to some aspects of the present disclosure.

[0026] FIG. **14** illustrates an example vibroacoustic device positioned within parallel slats of a frame of a bed, according to some aspects of the present disclosure.

[0027] FIG. **15** is a close up view of an example vibroacoustic device positioned within parallel slats of a frame of a bed, according to some aspects of the present disclosure.

[0028] FIG. **16** illustrates an example vibroacoustic device attaching to a slat of a frame of a bed, according to some aspects of the present disclosure.

[0029] FIG. **17** is another view of a crib incorporating an example vibroacoustic device, according to some aspects of the present disclosure.

[0030] FIG. **18** is an exploded view of an example remote control, according to some aspects of the present disclosure.

[0031] FIG. **19** is a front view of an example remote control, according to some aspects of the present disclosure.

[0032] FIG. **20** is a rear view of an example remote control, according to some aspects of the present disclosure.

[0033] FIG. **21** is a perspective view of an example remote control, according to some aspects of the present disclosure.

[0034] FIG. **22** is a perspective view of a battery housing of an example remote control, according to some aspects of the present disclosure.

[0035] FIG. **23** is a cross-section view through section B-B in FIG. **20** of an example remote control, according to some aspects of the present disclosure.

[0036] FIG. **24** illustrates an example remote control and an example dock, according to some aspects of the present disclosure.

[0037] FIG. **25** illustrates a user attaching an example remote control to an example dock, according to some aspects of the present disclosure.

[0038] FIG. **26** illustrates an example remote control attached to an example dock, according to some aspects of the present disclosure.

[0039] FIG. **27** illustrates an example remote control attached to an example dock, according to some aspects of the present disclosure.

[0040] FIG. **28** illustrates an example system including an example vibroacoustic device, according to some aspects of the present disclosure.

[0041] FIG. **29** is a cross sectional view of an example vibroacoustic device, according to some aspects of the present disclosure.

[0042] FIG. **30** illustrates an example remote control, according to some aspects of the present disclosure.

[0043] FIG. **31** illustrates an example remote control and an example dock, according to some aspects of the present disclosure.

[0044] FIG. **32** illustrates an example vibroacoustic device, according to some aspects of the present disclosure.

[0045] FIG. **33** is a cross-section view through section C-C in FIG. **32**, according to some aspects of the present disclosure.

[0046] FIG. **34** is a view of a stuffed animal incorporating a vibroacoustic device, according to

some aspects of the present disclosure.
[0047] FIG. **35** shows an operation frequency of vibroacoustic therapy as compared to other acoustic uses, according to some aspects of the present disclosure.
[0048] FIG. **36** illustrates example operating levels of an example vibroacoustic device, according to some aspects of the present disclosure.
[0049] FIG. **37** illustrates an example method for facilitating sleep and/or rest using a vibroacoustic device, according to some aspects of the present disclosure.

DETAILED DESCRIPTION

[0050] The following detailed description references specific embodiments of the subject disclosure and accompanying figures, including the respective best modes for carrying out each embodiment. It shall be understood that these illustrations are by way of example and not by way of limitation. Particular embodiments of a vibroacoustic device will now be described in greater detail with reference to the figures.
[0051] The present subject disclosure describes devices and techniques which promote rest and sleep by using vibroacoustic technology. The initial years of parenthood are a period when individuals strongly seek ways for maintaining enough sleep hours and high-quality sleep. The devices and techniques presented herein are geared towards optimization of the ambient environment for faster and deeper sleep by influencing biological internal rhythms. An example vibroacoustic system of this disclosure utilizes vibroacoustic frequencies to improve and promote sleep and/or rest. The vibroacoustic systems can be attached to sleeping areas, such as beds, cribs, and other rest areas. The vibroacoustic systems may have separate controls (e.g., using a remote control) for ease of access.
[0052] The present subject disclosure is based on ambient optimization where parents can turn their nursery room into a more soothing, perfect-to-sleep environment through sensational devices. However, the present subject disclosure is not limited to parents and children, but more broadly applicable to a wider demographic and age group. The present subject disclosure is mostly presented with respect to infants and small children, for sake of simplicity, but one of ordinary skill in the art would understand the broader applicability of the concept.
[0053] FIG. **1** illustrates an example vibroacoustic system **10**, according to some aspects of the present disclosure. Vibroacoustic system **10** can include a vibroacoustic device **100**, a remote control **200**, a dock **300**, and a charger **400**. Vibroacoustic device **100** is configured to generate vibroacoustic frequencies to promote and facilitate sleep and rest. Remote control **200** is operable to control operations of vibroacoustic device **100**. Dock **300** is configured to receive remote control **200**. Charger **400** can provide power to vibroacoustic device **100**, for example, by using a cable **402**.
[0054] FIG. **2** illustrates an example vibroacoustic device **100** and remote control **200**, according to some aspects of the present disclosure. Vibroacoustic device **100** is connected to a cable **402** to receive power. Additionally, remote control **200** can be used to control operations of vibroacoustic device **100**.
[0055] FIG. **3** is an exploded view of an example vibroacoustic device **100**, according to some aspects of the present disclosure. Vibroacoustic device **100** includes a housing **110**, a speaker **120**, a circuit board **130**, a vibroacoustic speaker **140**, an assembly cover plate **150**, a first rattle reduction layer **160**, a second rattle reduction layer **170**, and a top connection plate **180**.
[0056] Housing **110** can have an elongated shape and an expanded central portion defining a storage compartment **112**. Housing **110** may also be referred to as a shell. Storage compartment **112** is operable to contain speaker **120**, circuit board **130**, and vibroacoustic speaker **140** therein. A section of housing **110** for storage compartment **112** can have apertures **114** to allow acoustic waves to pass therethrough. Apertures **114** also provide cooling heat dissipation functionality for heat generated inside storage compartment **112**. Housing **110** may also have one or more fastening areas **116** to facilitate fastening of various components of vibroacoustic device **100** together.

[0057] Speaker **120** is configured to generate acoustic sound waves. Speaker **120** can be in communication with circuit board **130** to receive signals from circuit board **130**. For example, speaker **120** can generate acoustic sound waves for a lullaby when receiving a signal from circuit board **130**. As another example, speaker **120** can increase and/or decrease a volume (e.g., by generating higher/lower magnitude acoustic sound waves) in response to a signal from circuit board **130** associated with a volume control command.

[0058] Circuit board **130** is configured to receive, generate, process, and communicate signals to and/or from speaker **120** and/or vibroacoustic speaker **140**. Circuit board **130** is also configured to communicate with remote control **200** to send and/or receive signals from remote control **200**. For example, remote control **200** can send signals to circuit board **130** to increase a vibration intensity generated by vibroacoustic speaker **140**. Circuit board **130** can then generate and send a signal that causes vibroacoustic speaker **140** to increase the vibration intensity.

[0059] Vibroacoustic speaker **140** is configured to generate vibrations. As will be discussed in further detail below, vibroacoustic frequencies can provide various benefits and promote sleep and/or rest. Vibroacoustic speaker **140** can generate vibrations at varying levels of intensity, pattern, frequency, and the like. For example, vibroacoustic speaker **140** may generate stronger vibrations during one period of time and weaker vibrations during another period of time. It is to be understood that the vibroacoustic speaker **140** may also generate acoustic sound waves in addition to the generation of vibrations and may also produce characteristics similar to the speaker **120** as mentioned above.

[0060] Assembly cover plate **150** provides a cover for housing **110**. Housing **110** and assembly cover plate **150** cover storage compartment **112**, encasing speaker **120**, circuit board **130**, and vibroacoustic speaker **140** therein. Assembly cover plate **150** can include apertures **152** and a central aperture **154** to receive fasteners **192**, for example, to secure assembly cover plate **150** to housing **110**.

[0061] First rattle reduction layer **160** and second rattle reduction layer **170** are layers constructed of softer materials, which may be more efficient at absorbing noise and/or vibrations. These layers can also provide padding between different components (e.g., assembly cover plate **150** and top connection plate **180**).

[0062] Top connection plate **180** and housing **110** can encase speaker **120**, circuit board **130**, vibroacoustic speaker **140**, assembly cover plate **150**, first rattle reduction layer **160**, second rattle reduction layer **170**. Top connection plate **180** can also include bolt **194** configured to align, fasten and secure the components inside of top connection plate **180** and housing **110**. For example, bolt **194** can protrude through fastening areas (e.g., fastening area **116** of housing **110**). Additional fasteners can also be used to secure housing **110** with top connection plate **180**. For example, nut **190** can mate with bolt **194** to secure top connection plate **180** to housing **110**. For example, nut **190** can secure housing **110**, speaker **120**, circuit board **130**, vibroacoustic speaker **140**, assembly cover plate **150**, first rattle reduction layer **160**, an object for securing vibracoustic device **100** to (e.g., slats of a bed), second rattle reduction layer **170**, and top connection plate **180**. Fasteners may also be used internally. For example, fastener **192** can be used to secure assembly cover plate **150** to housing **110**.

[0063] FIG. **4** and FIG. **5** are a top view and a bottom view, respectively, of an example vibroacoustic device **100**, according to some aspects of the present disclosure. Vibroacoustic device **100** can include top connection plate **180** having a similar shape as an elongated portion of housing **110**. The elongated portion of housing **110** can include a first longitudinal end **110***a* and a second longitudinal end **110***b* at distal ends of housing **110**.

[0064] FIG. **6** and FIG. **7** are a lower perspective view and an upper perspective view, respectively, of an example vibroacoustic device **100**, according to some aspects of the present disclosure. A first side of vibroacoustic device **100** can include housing **110**, assembly cover plate **150**, contents stored within housing **110** and assembly cover plate **150** (e.g., speaker **120**, circuit board **130**, and/or vibroacoustic speaker stored in storage compartment **112**), and first rattle reduction layer **160**. A second side of vibroacoustic device **100** can include second rattle reduction layer **170** and top connection plate **180**. The first side and the second side can have a gap therebetween. The gap can be used, for example, to accommodate a surface for attaching vibroacoustic device **100** thereto. As will be discussed further below with respect to FIG. **16**, vibroacoustic device **100** can sandwich slats between the first side and the second side. Bolts **194** and fasteners **192** can secure the first side and the second side, while also providing enough space to maintain the gap.

[0065] FIG. **8** is a side view of an example vibroacoustic device **100**, according to some aspects of the present disclosure. In some embodiments, vibroacoustic device **100** can include various control features **118** including, but not limited to, an on/off switch, sound option switch, heat/cool feature, volume adjust, and vibration adjust, among others. Various methods may be used to control vibroacoustic device **100**, including control features **118** directly on the product, control remotely via Bluetooth/Infrared Light, control via smart device (phone, smartwatch, etc.), gesture/motion control, remote control **200**, sensors (e.g., motion sensors, weight sensors, etc.) and combinations thereof. Other methods may also be used, as appreciated by one having ordinary skill in the art.

[0066] FIG. **9** is a cross-section view through section A-A in FIG. **5** of an example vibroacoustic device **100**, according to some aspects of the present disclosure. Vibroacoustic device **100** is illustrated in a closed position with no gap between between the first side and the second side, in contrast to FIG. **6** and FIG. **7**. Instead, housing **110** encases, within storage compartment **112**, speaker **120**, circuit board **130**, and vibroacoustic speaker **140**. Assembly cover plate **150** is secured to housing **110** using aperture **152**, central aperture **154**, and fastener **192**. Housing **110**, assembly cover plate **150**, and the contents encased therewithin are secured to top connection plate **180**. More specifically, bolt **194** is attached to top connection plate **180** and received through corresponding holes on housing **110** (e.g., at fastening areas **116**) and assembly cover plate **150**.

[0067] FIG. **10** illustrates an example vibroacoustic device **100** attached to a crib **12**, according to some aspects of the present disclosure. Crib **12** may have a frame that secures a mattress, to which the vibroacoustic device **100** is attached.

[0068] FIG. **11** illustrates attachment of an example vibroacoustic device **100** to a crib **12**, according to some aspects of the present disclosure. The vibroacoustic device **100** according to the present subject disclosure is versatile in that it may be used in various ways. As depicted in FIG. **11**, vibroacoustic device **100** is connected to a standard crib **12**. A bottom portion of crib **12** may be a board, which may be wooden, plastic or other suitable material. Housing **110** of vibroacoustic device **100** can be attached to the board. For example, connectors (e.g., screws, bolts, fasteners, etc.) may be used to attach longitudinal ends **110***a*, **110***b* of housing **110** to the board. In some embodiments, the board may be positioned above a wire frame or may be attached to any part of the frame structure of the crib **12**, such as the side railing slats.

[0069] FIG. **12** and FIG. **13** illustrate an example vibroacoustic device **100** attached to a frame of a crib, according to some aspects of the present disclosure. FIG. **12** shows another method of attaching vibroacoustic device **100** to a crib **12**. Some such cribs **12** contain a wire frame **14**, which serves to support a mattress from underneath.

[0070] FIG. **13** shows a closer view of vibroacoustic device **100** attached to the crib **12**. The vibroacoustic device **100** may be inserted between vertical and/or horizontal wires within the frame **14** and locked in by extending the longitudinal ends **110***a*, **110***b* of housing **110** over a top side of the wires, while the narrower bottom portion (e.g., the portion including storage compartment **112**) extends downward and underneath the wire frame.

[0071] FIG. **14** and FIG. **15** illustrate an example vibroacoustic device **100** positioned within parallel slats **16** of a frame **14** of a bed, according to some aspects of the present disclosure. FIG. **15** shows a closer view of vibroacoustic device **100** positioned within the slats **16**. Vibroacoustic device **100** may be positioned in between two slats **16** such that the elongated portion of housing **110** is positioned above the top planar surface of two adjacent slats **16**, and the bottom portion (e.g., the portion including storage compartment **112**) hangs beneath the bottom surface of the two adjacent slats **16**. Positioning of the vibroacoustic device **100** in the manner shown involves several steps. First, the longitudinal ends **110***a*, **110***b* of the housing **110** are positioned such that they are parallel with the slats **16**. Next, the housing **110** or top portion of the vibroacoustic device **100** is lifted above the top surface of the slats **16**. Finally, vibroacoustic device **100** is turned such that the longitudinal ends **110***a*, **110***b* are positioned perpendicular to the parallel slats **16**, as shown in FIG. **14** and FIG. **15**. This position then "locks" the vibroacoustic device **100** in between the slats **16**.

[0072] FIG. **16** illustrates an example vibroacoustic device **100** attaching to a slat of a frame of a bed, according to some aspects of the present disclosure. More specifically, vibroacoustic device **100** can be attached to one or more slats **16** of a frame **14** of a bed or crib **12**. For example, first rattle reduction layer **160** and second rattle reduction layer **170** may sandwich rods or slats. For example, first rattle reduction layer **160** can be secured against assembly cover plate **150** and one side of the slats, while second rattle reduction layer **170** is secured against top connection plate **180** and another side of the slats. Consequently, a first side having housing **110**, assembly cover plate **150**, contents stored within housing **110** and assembly cover plate **150** (e.g., speaker **120**, circuit board **130**, and/or vibroacoustic speaker stored in storage compartment **112**), and first rattle reduction layer **160** attach to a second side having second rattle reduction layer **170** and top connection plate **180** and sandwiches slats between the first side and the second side. Vibroacoustic device **100** is thereby secured to the bed. First rattle reduction layer **160** and second rattle reduction layer **170** provide noise and vibration reduction and/or absorption, to prevent rattling of vibroacoustic device **100** against bed slats (or other object that vibroacoustic device **100** is secured to) during operation.

[0073] FIG. **17** is another view of a crib incorporating an example vibroacoustic device **100**, according to some aspects of the present disclosure. As illustrated, vibroacoustic device **100** provides acoustic sound **18** and vibration **20**. For example, vibroacoustic device **100** can provide audible soothing sound **18** and vibration **20** that correlate with the acoustic sound **18**. The audible sound may include some additional sounds such as a heartbeat sound and the mother's voice sound among other sounds.

[0074] FIG. **18** is an exploded view of an example remote control **200**, according to some aspects of the present disclosure. Remote control **200** can have a capsule-like shape. Remote control **200** can include an interface board **210**, a housing **220**, a button panel **230**, a magnet **240**, a circuit board **250**, an inner housing **260**, a power supply **270**, and a cover **280**.

[0075] Interface board **210** can be disposed on housing **220**. Interface board **210** provides a surface that can be easily removed for cleaning. In some embodiments, interface board **210** may have an aperture therein to provide access to a control (e.g., button panel **230**). Interface board **210** may be constructed such that interface board **210** flexes to press against and actuate button panel **230** when pressed. In some embodiments, housing **220** may have button actuators **222** between interface board **210** and button panel **230**, such that when interface board **210** is pressed and flexes against the button actuators **222**, the button actuators **222** press against button panel **230**.

[0076] Button panel **230** are configured to receive pressure, generate signals, and send the signals to circuit board **250**. Button panel **230** can include a center button **232** that comes in contact with a PCB button **254** in a center of circuit board **250**. In some embodiments, center button **232** can have a concave curvature to accommodate dock **300**. Button panel **230** can include buttons **234** that are configured to come into contact with interface board **210** when actuators **222** of interface board **210** are pressed. Buttons **234** can then press against and come in contact with PCB buttons **254** in corresponding positions.

[0077] Circuit board **250** is configured to receive signals from button panel **230**. Circuit board **250** can include an aperture **252** to facilitate proper positioning of circuit board **250** in relation to interface board **210**, housing **220**, button panel **230**, and inner housing **260**. For example, inner housing **260** can include a positioning protrusion **268** to be inserted into aperture **252**. Circuit board

250 can generate and send signals to another device (e.g., vibroacoustic device **100**, a mobile device, etc.). For example, a user may press a button **234** (e.g., by pressing against a corresponding portion of interface board **210**) that indicates the user wants to increase a vibration intensity. Circuit board **250** can receive the signal from the button **234** (e.g., from the contact of button **234** against PCB button **254**) and, in response to receiving the signal, generate and send a command signal to vibroacoustic device **100**, which can then increase the vibration intensity.

[0078] Referring back to magnet **240**, magnet **240** is operable to facilitate securing of remote control **200** to a dock **300**. In some embodiments, dock **300** may also be a charging dock configured to charge power supply **270**.

[0079] Inner housing **260** can provide a layer of protection between a power supply **270** and circuit board **250**. Additionally inner housing **260** provides an interface to allow power from power supply **270** to power circuit board **250**. Inner housing **260** can include a securing recess **262** to receive and secure a securing protrusion **282** of cover **280**. Inner housing **260** can also include a fastener aperture **264** to receive fastener **290** and thereby secure circuit board **250**. Inner housing **260** can also include a battery housing **266**. For example, in some embodiments, power supply **270** can be a battery and battery housing **266** can be configured to secure the battery and direct power from the battery to the circuit board **250**. Fasteners **290**, **292** can be used with inner housing **260** to secure components together. For example, fastener **290** can fasten buttons with circuit board **250** and fastener **292** can fasten inner housing **260** with housing **220**.

[0080] Cover **280** and housing **220** encase button panel **230**, magnet **240**, circuit board **250**, inner housing **260**, and power supply **270**. Cover **280** can include a securing protrusion **282** configured to mate with securing recess **262** to secure cover **280** to inner housing **260**. In some embodiments, securing protrusion **282** can be configured to abut against a positioning shoulder **269** (obscured from view) of inner housing **260**. Cover **280** can also include a securing recess **284** to receive and secure a securing protrusion (obscured from view) of inner housing **260**. Cover **280** can also include guides **286** to provide a more controlled, longitudinal movement of cover **280** in relation to inner housing **260** and housing **220**. For example, cover **280** can slide longitudinally against inner housing **260** to either secure and/or remove cover **280** from inner housing **260** and housing **220**.

[0081] FIG. **19** is a front view of an example remote control **200**, according to some aspects of the present disclosure. As shown, interface board **210** and center button **232** are visible and can receive input from a user.

[0082] FIG. **20** is a rear view of an example remote control **200**, according to some aspects of the present disclosure. In the closed position, cover **280** covers the rear face of remote control **200**.

[0083] FIG. **21** is a perspective view of an example remote control **200**, according to some aspects of the present disclosure. More specifically, FIG. **21** illustrates cover **280** in a semi-open position, such that cover **280** is sliding along housing **220** and inner housing **260**. Cover **280** can slide downwards to cover inner housing **260** and/or slide upwards to reveal inner housing **260**.

[0084] FIG. **22** is a perspective view of a battery housing of an example remote control **200**, according to some aspects of the present disclosure. When cover **280** is removed from housing **220** and inner housing **260**, battery housing **266** is accessible and a power supply **270** (e.g., a battery) can be placed therein.

[0085] FIG. **23** is a cross-section view through section B-B in FIG. **20** of an example remote control, according to some aspects of the present disclosure. Remote control **200** is depicted facing down, such that the front (e.g., the interface board **210**) of the remote control **200** faces downwards. Interface board **210** sits within a recess of housing **220** and is contact with buttons **234**. Additionally, center button **232** sits within a center of interface board **210**. Center button **232** also may have a concave curvature that matches a corresponding convex curvature of a dock **300** (e.g., mounting surface **310** of dock **300**). Aperture **252** of circuit board **250** receives positioning protrusion **268** of inner housing **260** to properly position circuit board **250** within inner housing **260** and housing **220** to ensure contact of center button **232** and buttons **234** with PCB buttons **254**, such that pressing center button **232** and/or buttons **234** (e.g., by pressing interface board **210** against actuators **222**, which actuates button **234**) activates PCB buttons **254**.

[0086] When cover **280** is in the closed position (e.g., covering inner housing **260**), one securing protrusion **282** can be positioned away from positioning shoulder **269**, while another securing protrusion **282** can be positioned within securing recess **262** of inner housing **260**. When removing cover **280**, cover **280** slides leftwards (with respect to FIG. **23**) to free the leftmost securing protrusion **282** from securing recess **262**, while causing the rightmost securing protrusion **282** to abut against positioning shoulder **269**. When the rightmost securing protrusion **282** abuts against positioning shoulder **269**, cover **280** can be pulled upwards and/or away from inner housing **260** to be removed. In some embodiments, Cover **280** can have a curvature to provide space for power supply **270**, which is secured by battery housing **266**.

[0087] FIG. **24** illustrates an example remote control **200** and an example dock **300**, according to some aspects of the present disclosure. As discussed above, remote control **200** can have center button **232**, which can have a curvature that corresponds to a curvature of mounting surface **310**. Mounting surface **310** can also include a magnet disposed therein. Mounting surface **310** can receive and secure remote control **200** using the magnet disposed in mounting surface **310** and magnet **240** disposed adjacent to center button **232**. Remote control **200** can also include connector **224** configured to mate with connector **320**. Connectors **224**, **320** can allow dock **300** to charge remote control **200**. In some embodiments, dock **300** may include a power cable **330**.

[0088] FIG. **25** illustrates a user attaching an example remote control **200** to an example dock **300**, according to some aspects of the present disclosure.

[0089] FIG. **26** illustrates an example remote control **200** attached to an example dock **300**, according to some aspects of the present disclosure. In some embodiments, remote control **200** can include an indicator **288**, such as an light emitting diode (LED) to provide additional information (e.g., remaining power, current mode, etc.).

[0090] FIG. **27** illustrates an example remote control **200** attached to an example dock **300**, according to some aspects of the present disclosure. Dock **300** can be placed in various different places across a home, such as on a wall near a light switch for ease of access.

[0091] FIG. **28** illustrates another exemplary embodiment including an example vibroacoustic device **1100**, according to some aspects of the present disclosure. FIG. **28** shows an exemplary vibroacoustic device **1100** with separate controls control interface **1200** and a power supply **1400** (e.g., a standard AC wall adapter), in communication through various wire connections. The control interface **1200** includes controls **1230** in the form of various buttons and switches. In this particular exemplary embodiment, the buttons and switches are separate from the body of the vibroacoustic device **1100**.

[0092] FIG. **29** is a cross sectional view of an example vibroacoustic device **2100**, according to some aspects of the present disclosure. The vibroacoustic device **2100** has a housing **110** having an elongated and wider top portion with longitudinal ends **2110*a***, **2110*b*** and a narrower bottom portion for a storage compartment **2112**. The top side of the housing **2110** can include a solid material (such as wood) assembly cover **2150** to more securely fasten to a bed slat, as described above. Alternatively, the top portion and assembly cover **2150** can be a two-piece separable construction in which the assembly cover **2150** becomes a top plate and is separated from the housing **2110** and placed on one side of an object and the assembly cover **2150** is placed on another side of the object and both the housing **2110** and the assembly cover **2150** are connected to each other to secure the vibroacoustic device **2100** to the object as discussed above. A vibroacoustic speaker **2140**, which generates sound and vibration, is positioned within the housing **2110** of the vibroacoustic device **2100** directly underneath the assembly cover **2150**. A printed circuit board (PCB) **2130** is positioned underneath the vibroacoustic speaker **2140**, and includes the various electronic control tools, circuitry, modules, memories, and sound required to operate the vibroacoustic speaker **2140**. Housing **2110** can also include apertures **2114** to allow for dissipation of heat generated by PCB **2130**, and also provide further sound distribution.

[0093] FIG. **30** illustrates an example remote control **2200**, according to some aspects of the present disclosure. Remote control **2200** can have a different interface board **2210** having a different layout of buttons **2212**.

[0094] FIG. **31** illustrates an example remote control **3200** and an example dock **300**, according to some aspects of the present disclosure. Remote control **3200** can also have a different interface board **3210** having yet another layout of buttons **3212**. Additionally, various different layouts can be used for remote controls **200**, **2200**, **3200**, while also having a curvature to attach to dock **300**.

[0095] FIG. **32** illustrates an example vibroacoustic device **3100**, according to some aspects of the present disclosure. FIG. **33** is a cross-section view through section C-C in FIG. **32**, according to some aspects of the present disclosure. The vibroacoustic device **3100** should be accessible to make repairs, and/or be able to change or charge its internal battery. The control buttons **3118** and switches on housing **3110** can be similar to those shown in FIG. **8** and are operable to communicate with PCB **3130** to control vibroacoustic speaker **3140**. Housing **3110** can also have apertures **3114** to allow heat dissipation and sound to flow therethrough.

[0096] FIG. **34** is a view of a stuffed animal incorporating a vibroacoustic device **4100**, according to some aspects of the present disclosure. While, FIG. **34** illustrates a stuffed animal as an example, other products may also be incorporated with the disclosed vibroacoustic technology. For example, the disclosed vibroacoustic technology can be incorporated in stuffed dolls/animals, pillows, nursery pillows, body pillows, swings, bouncers, strollers, car seats, changers, pads, mattresses, blankets, clothes, among others. In some examples, the vibroacoustic device **4100** may be the product or may be positioned within the interior of the product (e.g., inside the stuffed animal) or a designated pocket on the product (e.g., blanket). Additionally, the present vibroacoustic technology can also be used in other products including, but not limited to, bassinets, cribs, beds, soft goods (e.g., clothing, swaddles, blankets, pillow covers, bed covers, chairs, sofas, baby loungers, etc.), bathtubs and related accessories, strollers, jumpers, swings, baby carriers, charging pads/stations, car seats, toys, body pillows, massagers, etc.

[0097] FIG. **35** shows an operation frequency of vibroacoustic therapy as compared to other acoustic uses, according to some aspects of the present disclosure.

[0098] As shown in the sound frequency scale **22** of FIG. **35**, vibroacoustic therapy involves sound therapy that uses audible sound vibrations. This reduces anxiety symptoms, supports sleep, invokes relaxation, and alleviates stress. The process works by passing low frequency sine wave vibrations into the body, which is believed to have deeper physiological and emotional effects compared to just gentle vibrations.

[0099] Vibroacoustic therapy is different from gentle vibration because it involves delivering sound waves and vibrations simultaneously. This dual sensory stimulation is believed to have deeper physiological and emotional effects compared to just gentle vibration. Such dual sensory delivery is thought to stimulate the nervous system, promote relaxation, and potentially relieve pain or anxiety.

[0100] Vibroacoustic therapy is effective because sound penetrates deeper—vibroacoustic therapy provides massage therapy to muscles and joints that hand/mechanical massage cannot reach. Ultrasound is a well-known and accepted sound technology for viewing tissue inside a body and it operates at 20 KHz+. Meanwhile, as shown in FIG. **35**, vibroacoustic therapy operates in the acoustic region, which is 20 Hz to 20 KHz. More specifically, vibroacoustic therapy operates closer to 20 Hz.

[0101] FIG. **36** illustrates example operating levels of an example vibroacoustic device **100**, according to some aspects of the present disclosure. Vibroacoustic device **100** is configured to operate in various different modes. For example, a first mode can be designed to facilitate falling asleep. The first mode can include highest level of vibration intensity, while also including a minimal amount of acoustic noise. Vibroacoustic device **100** and transition to a second mode by gradually increasing and/or decreasing vibrations and/or acoustic noise. A second mode can be designed to maintain sleep by stopping vibrations and increasing acoustic noise (e.g., pink noise). Modes can be set for pre-determined durations. For example, when a child wakes up in the middle of the night, a user can press a button on remote control **200** to set vibroacoustic device **100** in the first mode to facilitate the child in falling back asleep. After the pre-determined duration (e.g., when the child has likely fallen asleep), the vibroacoustic device can return to the second mode to keep the child asleep.

[0102] In addition to the path shown in FIG. **36**, it is to be understood that another aspect of this invention is the ability for the vibroacoustic device **100** to coordinate and overlay acoustic sounds and vibrations at various vibration and acoustic sound levels. For example, during the first “GO-TO-SLEEP” phase a first vibration from the vibroacoustic speaker **140** may mimic the heartbeat of the mother and can be produced at a first vibrational frequency. Simultaneously, a first sound from the speaker **120** such as a lullaby or pink noise can be produced at a first acoustic sound frequency. The first phase can be programmed at any period of time, for example 5, 15, 30 or 60 mins. As the phase progresses, the vibrational frequency of the mothers heartbeat can be reduced and slowed to mimic the mother's body moving into a sleep state. Likewise, the first acoustic sound frequency of the first sound can also be reduced to coincide with the reduction and slowing down of the mothers heartbeat so that when the “KEEP-ASLEEP” phase has begun, the first sound may be substantially low and/or completely eliminated. If, in the third phase “BACK-TO-SLEEP”, the infant wakes up, the initial vibration and acoustic level can be restored and eventually lowered to encourage the infant to go back to sleep. Another phase named “WAKE-UP” can be included which reverses the “GO-TO-SLEEP” phase to gradually wake up the sleeping infant when it is time to gently wake up the infant. It is to be understood that numerous routines are possible in the various phases and that the vibrational frequency of the vibroacoustic speaker **140** and the acoustic sound levels of the speaker **120** may be coordinated and overlayed at various preferred vibration and acoustic sound levels.

[0103] FIG. **37** illustrates an example method for facilitating sleep and/or rest using a vibroacoustic device, according to some aspects of the present disclosure. Although the example method **3700** depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the method **3700**. In other examples, different components of an example device or system that implements the method **3700** may perform functions at substantially the same time or in a specific sequence.

[0104] At operation **3710**, a vibroacoustic device (e.g., vibroacoustic device **100**) generates a first combination of vibroacoustics and noise. For example, the vibroacoustic device can generate a higher level of vibroacoustic vibrations and a relatively low level of noise to facilitate falling asleep.

[0105] At operation **3720**, the vibroacoustic device generates a transition combination of vibroacoustics and noise (e.g., after a predetermined amount of time). The transition combination can be a combination that is a smooth transition from the first combination to a second combination of vibroacoustics and noise. For example, the vibroacoustic device can generate a decreasing intensity of vibroacoustic vibrations from the higher level of vibroacoustic vibrations to a low level or absence of vibroacoustic vibrations and an increasing level of pink noise from the relatively low level of noise to a higher level of pink noise.

[0106] At operation **3730**, the vibroacoustic device generates the second combination of vibroacoustics and noise. For example, the vibroacoustic device can generate an intensity of vibroacoustic vibrations that is lower than the higher level of vibroacoustic vibrations and a level of pink noise that is higher than the relatively low level of noise of the first combination.

[0107] At operation **3740**, the vibroacoustic device determines that a user (e.g., a baby on the crib that the vibroacoustic device is attached to) has woken from sleep. For example, the user may toss and turn, cry, and/or perform other actions that are indicative of waking from sleep. In some embodiments, the vibroacoustic device can have sensors that are configured to capture data that is associated with these actions and determine, based on the captured data, that the user has woken from sleep. In some embodiments, a user may generate a command to the vibroacoustic device (e.g., using a remote control such as remote control **200**) to generate vibroacoustics and noise, thereby indicating that the user has woken.

[0108] At operation **3750**, the vibroacoustic device generates a second transition combination of vibroacoustics and noise. The second transition combination can be a combination that is a smooth transition from the second combination back to the first combination of vibroacoustics and noise. For example, the vibroacoustic device can generate an increasing intensity of vibroacoustic vibrations from the low level or absence of vibroacoustics vibrations to the higher level of vibroacoustic vibrations and a decreasing level of pink noise from the higher level of pink noise to the relatively lower level of noise.

[0109] At operation **3760**, the vibroacoustic device generates the first combination of vibroacoustics and noise. For example, the vibroacoustic device can generate the higher level of vibroacoustic vibrations and the relatively lower level of noise to facilitate falling asleep.

[0110] At operation **3770**, the vibroacoustic device generates the transition combination of vibroacoustics and noise (e.g., after a predetermined amount of time). For example, the vibroacoustic device can generate a decreasing intensity of vibroacoustic vibrations from the higher level of vibroacoustic vibrations to a low level or absence of vibroacoustic vibrations and an increasing level of pink noise from the relatively low level of noise to a higher level of pink noise.

[0111] At operation **3780**, the vibroacoustic device generates the second combination of vibroacoustics and noise. For example, the vibroacoustic device can generate an increasing intensity of vibroacoustic vibrations from the low level or absence of vibroacoustics vibrations to the higher level of vibroacoustic vibrations and a decreasing level of pink noise from the higher level of pink noise to the relatively lower level of noise.

[0112] At operation **3790**, the vibroacoustic device generates a third transition combination of vibroacoustics and noise (e.g., after a predetermined amount of time). The third transition combination can be a combination that is a smooth transition from the first and/or second combinations to an inactive status. For example, the third transition combination can decrease vibroacoustic vibrations and/or noise levels until the vibroacoustic device stops generating vibroacoustic vibrations and/or noise.

[0113] The illustrations and examples provided herein are for explanatory purposes and are not intended to limit the scope of the appended claims. It will be recognized by those skilled in the art that changes, or modifications may be made to the above-described embodiments without departing from the broad inventive concepts of the invention. It is understood therefore that the invention is not limited to the particular embodiments which are described but is intended to cover all modifications and changes within the scope and spirit of the subject disclosure.

[0114] The foregoing disclosure of the exemplary embodiments of the present subject disclosure has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the subject disclosure to the precise forms disclosed. Many variations and modifications of the embodiments described herein will be apparent to one of ordinary skill in the art in light of the above disclosure. The scope of the subject disclosure is to be defined only by the claims appended hereto, and by their equivalents.

[0115] Further, in describing representative embodiments of the present subject disclosure, the specification may have presented the method and/or process of the present subject disclosure as a particular sequence of steps. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the specification should not be construed as limitations on the claims. In addition, the claims directed to the method and/or process of the present subject disclosure should not be limited to the performance of their steps in the order written, and one skilled in the art can readily appreciate that the sequences may be varied and still remain within the spirit and scope of the present subject disclosure.

Claims

1. A vibroacoustic system, comprising: a vibroacoustic device having: a housing, a vibroacoustic speaker disposed in the housing, the vibroacoustic speaker configured to generate vibrations, and an acoustic speaker disposed in the housing, the acoustic speaker configured to generate acoustic sounds; and control features configured to control the vibroacoustic device, wherein the control features cause the vibroacoustic device to generate coordinated vibrations and acoustic sounds.
2. The vibroacoustic system of claim 1, wherein the vibroacoustic device is configured to generate a first combination of vibrations and acoustic sounds and a second combination of vibrations and acoustic sounds, and wherein the first combination includes a higher level of vibrations and a lower level of noise compared to a lower level of vibrations and a higher level of noise of the second combination.
3. The vibroacoustic system of claim 1, wherein the vibroacoustic device is configured to generate a transition combination of vibrations and acoustic sounds to transition between a first combination of vibrations and acoustic sounds and a second combination of vibrations and acoustic sounds.
4. The vibroacoustic system of claim 1, wherein the housing is adapted to be positioned between slats.
5. The vibroacoustic system of claim 1, wherein the vibroacoustic device further includes a connection plate, wherein the connection plate is operable to be secured to the housing.
6. The vibroacoustic system of claim 5, wherein the connection plate and the housing are operable to secure slats therebetween.
7. The vibroacoustic system of claim 5, wherein the housing has an elongated portion, and wherein the connection plate is secured to longitudinal ends of the housing with fasteners, and wherein the housing and the connection plate sandwich a slat.
8. The vibroacoustic system of claim 1, wherein the controls features are disposed on a remote control that is configured to communicate with the vibroacoustic device.
9. The vibroacoustic system of claim 8, further comprising: a dock operable to receive and secure the remote control.
10. The vibroacoustic system of claim 1, wherein the controls features are disposed on the housing.
11. The vibroacoustic system of claim 1, wherein the vibroacoustic device further includes a printed circuit board (PCB) disposed within the housing, and wherein the PCB receives signals from the control features and causes the vibroacoustic speaker and the acoustic speaker to generate the vibrations and the acoustic sounds.
12. The vibroacoustic system of claim 1, wherein the vibrations mimic a heartbeat pattern.
13. The vibroacoustic system of claim 1, wherein the vibrations and acoustic sounds are overlayed over each other.
14. A vibroacoustic device, comprising: a housing; a vibroacoustic speaker disposed in the housing, the vibroacoustic speaker configured to generate vibrations; an acoustic speaker disposed in the housing, the acoustic speaker configured to generate acoustic sound; and a printed circuit board (PCB) in communication with the vibroacoustic speaker and the acoustic speaker, wherein the PCB causes the vibroacoustic speaker and the acoustic speaker to generate coordinated vibrations and acoustic sounds.
15. The vibroacoustic device of claim 14, wherein the vibroacoustic speaker and the acoustic speaker are configured to generate a first combination of vibrations and acoustic sounds and a second combination of vibrations and acoustic sounds, and wherein the first combination includes a

higher level of vibrations and a lower level of noise compared to a lower level of vibrations and a higher level of noise of the second combination.

16. The vibroacoustic device of claim 14, wherein the vibroacoustic speaker and the acoustic speaker are configured to generate a transition combination of vibrations and acoustic sounds to transition between a first combination of vibrations and acoustic sounds and a second combination of vibrations and acoustic sounds.

17. The vibroacoustic device of claim 14, wherein the PCB is in communication with control features to receive commands to control the vibroacoustic speaker and the acoustic speaker.

18. The vibroacoustic device of claim 17, wherein the control features are disposed on a remote control that is configured to communicate with the PCB.

19. A method comprising: generating, using a vibroacoustic device having a vibroacoustic speaker and an acoustic speaker, a first combination of vibrations and acoustic noises; generating, using the vibroacoustic device, a transition combination of vibrations and acoustic noises, wherein the transition combination is a gradual transition of intensity of vibrations and volume of acoustic noises from a first intensity and volume of the first combination to a second intensity and volume of a second combination of vibrations and acoustic noises; and generating, using the vibroacoustic device, the second combination of vibrations and acoustic noises.

20. The method of claim 19, further comprising: generating, using the vibroacoustic device, a second transition combination of vibrations and acoustic noises, wherein the second transition combination is a gradual transition of intensity of vibrations and volume of acoustic noises from the second intensity and volume of the second combination to the first intensity and volume of the first combination.